

Basics of Symmetric and Public Key Cryptography

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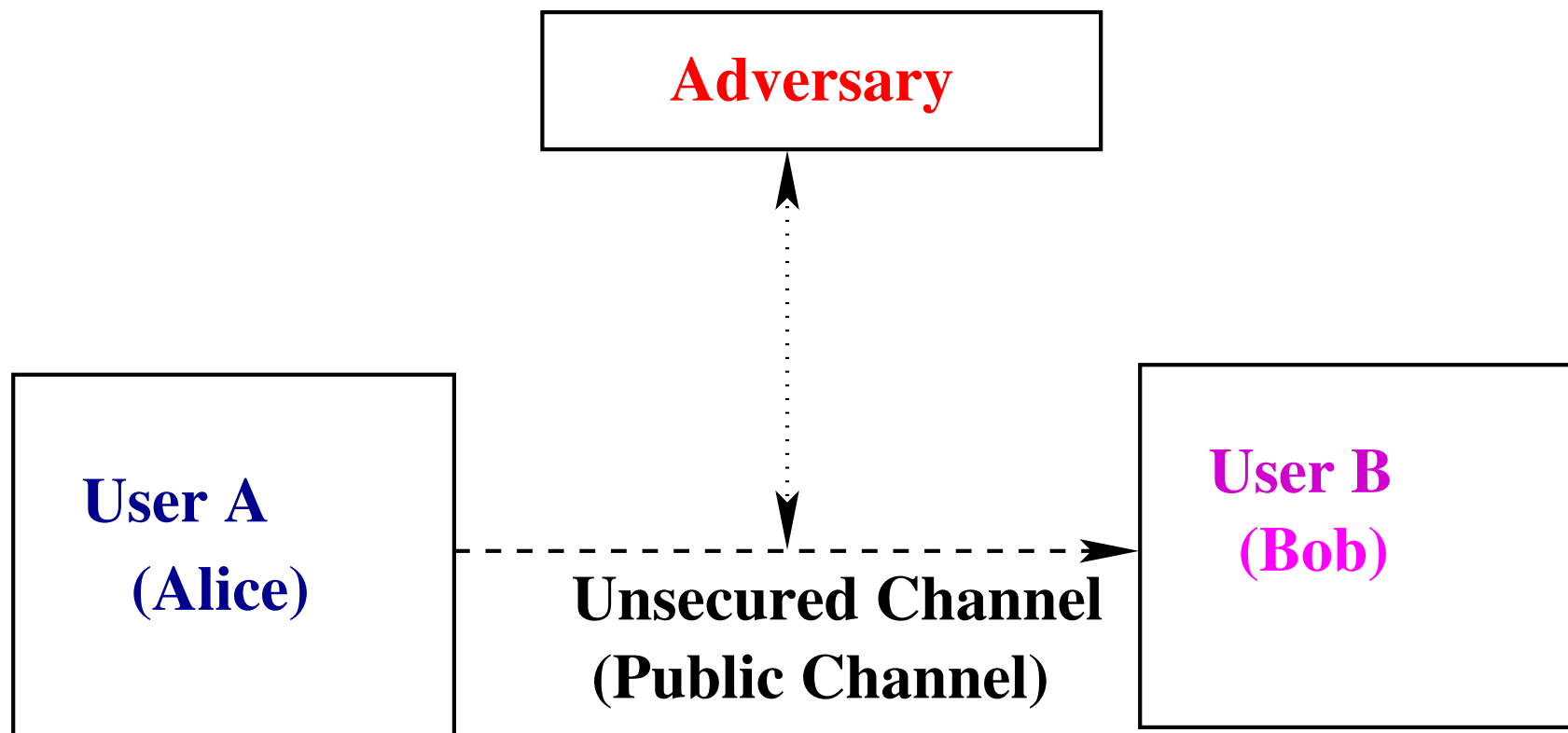
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What is Cryptography?

- Cryptography is the study of **mathematical techniques** related to aspects of information security such as confidentiality, data integrity, entity authentication, message authentication (data origin authentication) and non-repudiation.

Consider the following simple two-party communication model:



- An “**adversary**” is an entity in a two-party communication which is neither the sender nor the receiver, and which tries to defeat the information security service being provided between the sender and the receiver.
- A “**channel**” is a means of conveying information from one entity to another entity.
- An “**unsecured (public) channel**” is one from which parties other than the sender and the receiver can reorder, delete, insert, or read the data being transmitted.
- A “**secured channel**” is one from which an adversary does not have the ability to reorder, delete, insert, or read the data being transmitted.

Types of adversary

- A “**passive adversary**” is an adversary who is only capable of reading information from an unsecured channel.
- An “**active adversary**” is an adversary who is capable to transmit, alter, or delete information on an unsecured channel.

Cryptographic goals (objectives)

- **Confidentiality:** Privacy (confidentiality) is a service of keeping information secret from all but those who are authorized to see it.
- **Data integrity:** ensuring information has not been altered by unauthorized or unknown means.
- **Entity authentication or identification:** Corroboration of the identity of an entity (i.e., a person, a computer terminal, a credit card, etc.).
- **Message or data origin authentication:** Corroborating the source of information.
- **Non-repudiation:** Preventing the denial of the previous session (preventing the malicious nodes to hide their activities).

Cryptographic goals (objectives)

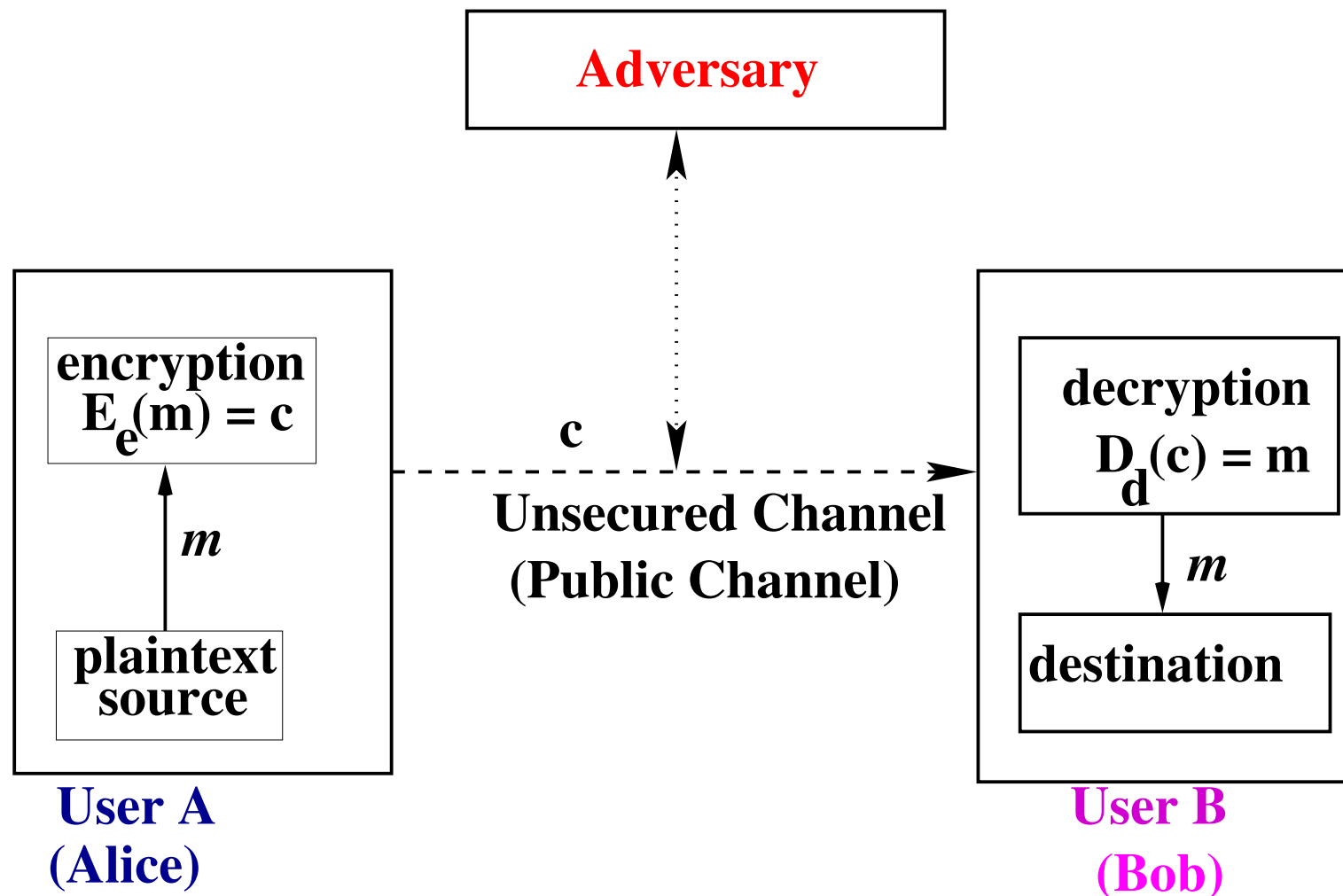
- **Authorization:** Conveyance to another entity such as a person or group of users. It ensures that the nodes (users) those who are authorized can be involved in providing information to network services.
- **Signature:** a means to bind information to an entity.
- **Access control:** restricting access to resources to privileged entity.
- **Certification:** endorsement of information by a trusted entity.

We need also to consider the forward and backward secrecy when new nodes join in the network and existing nodes depart from the network.

- **Forward secrecy:** When a node (user) leaves the network, it must not read any future messages after its departure.
- **Backward secrecy:** When a new node (user) joins in the network, it must not read any previously transmitted message.

Introduction to Cryptography

Consider the following simple two-party communication model with encryption:



- **Security of the scheme**

- ▶ Depends entirely on the secrecy of the key
- ▶ Does not depend on the secrecy of the algorithm (Needs to be public for criticism!)

- Hence, we make the **assumptions** as follows:

- ▶ Algorithms for encryption/decryption are known to the public
- ▶ Keys used are kept secret

Definition

An encryption scheme (cipher or cryptosystem) is said to be **breakable** if a third party, without prior knowledge of the key pair (e, d) , can systematically recover plaintext from corresponding ciphertext within some appropriate time frame.

Goal: We want this problem for an adversary (attacker) to be NP-hard (Computationally infeasible).

Definition (Brute-force attack)

An encryption scheme can be broken by trying all possible keys to see which one the communicating parties are using (assuming that the class of encryption functions is public knowledge). This is called an exhaustive search of the key space.

What is meant by “Security lies in the keys” (using brute-force attack)

Key Size (bits)	Number of Alternative Keys	Time Required at 1 Decryption/ μ s	Time Required at 10^6 Decryptions/ μ s
32	$2^{32} = 4.3 \times 10^9$	$2^{31} \mu\text{s} = 35.8 \text{ minutes}$	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55} \mu\text{s} = 1142 \text{ years}$	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127} \mu\text{s} = 5.4 \times 10^{24} \text{ years}$	$5.4 \times 10^{18} \text{ years}$
168	$2^{168} = 3.7 \times 10^{50}$	$2^{167} \mu\text{s} = 5.9 \times 10^{36} \text{ years}$	$5.9 \times 10^{30} \text{ years}$
26 characters (permutation)	$26! = 4 \times 10^{26}$	$2 \times 10^{26} \mu\text{s} = 6.4 \times 10^{12} \text{ years}$	$6.4 \times 10^6 \text{ years}$